MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10545705

(SIGHT CORRECTION)

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-5078S comprise the complete requirements for the sight Correction Circuit Card Assembly.

2. APPLICABLE DocUMENTS

2.1 The following documents of the issue in effect on the date of invitation forbids or request for proposal form •part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-50785	Circuit Card Assemblies, General Specification for

DRAWINGS

U.S. Army, Frankford Arsenal

1054S42S	Sight Current Generator
10545455	Receptacle
10545705	Circuit Card Assembly, Sight Connection

PACKAGING Data Sheet

10545705	Packaging of Circuit Card Assembly, Sight Correction

(Copies of specifications, standards, drawings, and publications requiredby Suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directedby the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:38Z
Check the source to verify that this is the currentversion before use.

REQUIREMENTS

3.1 item definition. This assembly is •single-ended printed circuit board consisting *of* 4 operational amplifiers, •voltage divider, •transistor switch, and •diode function generator. When installed in the Sight Current Generator Drawing number 1054S425, the assembly is used to invert and amplify the signals which *make* up *the* sight correction portion of the torque current voltage.

3.1.1 major component list. The compments of this asseably shall be as specified on the Engineering Parts List (EPL) 2054S705.

3.1.2 Government furnished property. Not applicable.

3.2 Characteristics

3.2.1 Performance. The assembly shall be capable of performhg *the* following requirments when assembled in accordance with Drawing Number 10545705.

a. Operational amplifier Al shall sum, amplify and invert the inputs required to develop the 2 signal.

b. Operstionsl eaplifier A_2 shall invent one of the outputs required to develop the torque current voltage.

c. Operational amplifier A3 shall invert *one of the inputs* required *to* develop the torque current voltage.

d. Operational amplifier a4 shall modify and invert the voltage developed by the diode function generator (CR4 through CR 8)

e. Transistor switch Q7 shall ground the input to the operational *amplifier* A4, when activated by the acquisition time delay.

f. The voltage divider R22, R23 shall develop the 13 Vdc *reference* signal required in the "Ready to Fire" light logic circuitry.

3.2.1.1 Input and output levels

3.2.1.1.1 Input power. The inpt *power* supplied shall be as specified in Table I.

Tsble X. Input *Power*

Pin	Voltage
4 and 25	+15 ± 2.0 percent Vdc
14	-15 ± 2.0 percent Vdc
1 and 28	Ground

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:38Z
Check the source to verify that this is the current version before use.

3.2.1.1.2 output levels. The requirements specified in 3.2.1.1.2.1 through 3.2.1.1.2.6 shall be met when supplied the input voltages specified in Table I.

3.2.1.1.2.1 Offset voltage. The offset voltage *of each* operational amplifier shall be zero Vdc ± 50 millivolts with zero Vdc applied to the input pins. The input and output shall be as specified in Table II.

Table II. Amplifier **Pins**

Amplifier	Input Pins	Output Pins
Al	5 and 6	7
A2	8	9
A3	26	19
A4	27	10

3.2.1.1.2.2 Pin 7 output. The output shall be [Vin (-0.8666~ ± 1.0 percent between pin 7 and pin one with an input voltage of -15 Vdc to zero Vdc applied to either pin 5 or pin 6.

3.2.1.1.2.3 Pin 9 output. The output shall be [Vin (-1.0)] ± 1.0 percent between pin 9 and pin one with an input voltage of -13 Vdc to +13 Vdc applied to pin 8.

3.2.1.1.2.4 Pin 10 output. The output shall be as specified in •. and b. between pin 10 and pin one.

a. $\left[-\frac{(Vin)^2}{13}\right]$ **± 1.0 percent with an input voltage of zero Vdc to +13** applied to pin 27 and zero Vdc applied to pin 12.

b. Zero Vdc ± 50 millivolts with an input voltage of zero Vdc to +13 Vdc applied to pin 25 and +15 Vdc ●*pplied* to pin 12.

3.2.1.1.2.5 Pin 19 output. The output shall be *[Vin* (-1.0)] ± 1.0 percent between pin 19 and pin 28 with an input voltage of -13 Vdc to +13 Vdc applied to pin 26.

3.2.1.1.2.6 Pin 23 output. The ratio of the output between pin 23 and pin 28 to the input between pin 4 and pin one shall be .8667 ± 0.2 percent.

3.2.1.2 Environmental. The assembly shall *meet the* environmental requirments of specification MIL-C-50785.

3.2.2 Physical characteristic. The physical characteristics shall be as specified in drawing 10545705 and the engineering parts list.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:38Z
Check the source to verify that this is the current version before use.

3.3 Desire and construction

3.3.1 Production drawings . This assembly shall be *fabricated in accordance with the requirements* of specification MIL-C-S078S, drawing 1054S705 and *associated* lists .

3.3.2 Standards of manufactured. The standards of manufacturee shall be those specified in specification MIL-C-50785

3.3.3 Workmanship. The workmanship shall be as *specified in* specification MIL-C-S0785.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:38Z
Check the source to verify that this is the current version before use.

4. QUALIIY ASSURANCE PROVISION

4.1 General

4.1.1 Responsibility for inspection. Thee responsibility for inspection shall be as specified in specification MIL-C-50785.

4.1.2 Special tests and examination. The assembly shaIl undergo first article and control testing in accordance with 4.1.2 Of Specification MIL-C-50785.

4.2 Quality conformance inspections. The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-50785.

4.2.1 Test equipment

4.2.1.1 Power supply (pSI). The power supply shall be 15 ± 0.1 Vdc, 100 milliamperes (ma) or *greater,* ripple less than 10 millivolts (mv) rms or 100 mv peak-to-peak.

4.2.1.2 Power SUPPLY (pS2). The power supply shall be -15 ± 0.1 Vdc, 100ma *or* greater, ripple less than 10 mv *rms* 100 mv peak-to-peak.

4.2.1.3 Power SUPPIy [PS ~. The power supply shall be zero ± 15 Vdc, with accuracy of adjustment and regulation to ± .01 Percent, 10 ma or greater.

4.2.1.4 Digital voltmeter. The digital voltmeter shall have • voltage • ccuracy and resolution *of* 0.1 percent or better from zero to 100 vdc. Input impdancee greater than 10 megohms on • ll voltage scales.

4.2.2 Functional tests

4.2.2.1 Test *connections.* *'fhe per* supplies shall be connected to the assembly under test in accordance with Figure 1 using receptacle 105454S5.

4.2.2.2 Test procedure. Compliane. with the requlrements *of* 3.2.1.1 through 3.2.1.1.6 shall be verified as follows:

• . Power supply (PSS) voltage shall be adjusted and connected to the specified input pin *per* Table III. A digital voltmeter shall be used to measure the voltage at tho appliable pin to the limits indicated in Table 111.

b. The ratio of the voltage on pin 23 to the voltage on pin 4 shall be .8667 ± 0.2 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:38Z
Check the source to verify that this is the current version before use.

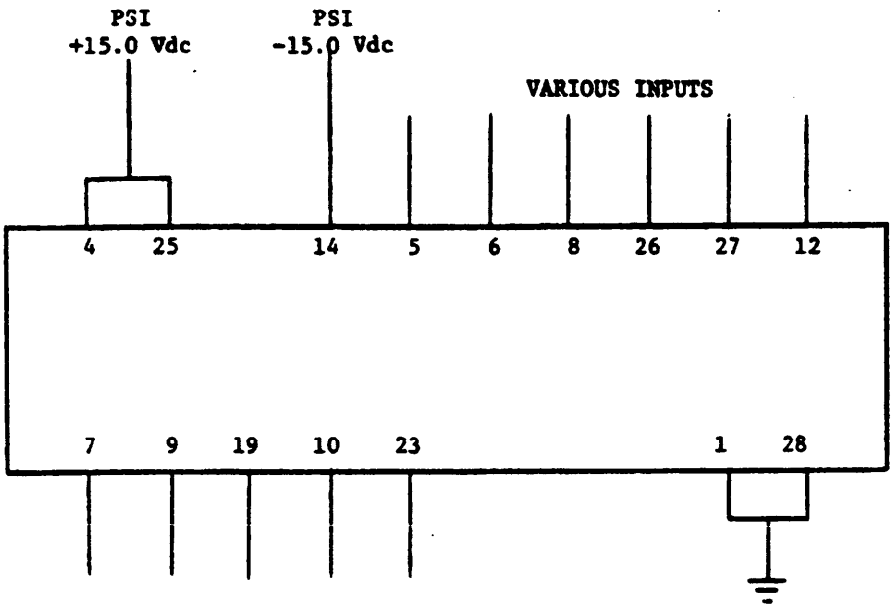


Figure 1. Test Connections

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:38Z
Check the source to verify that this is the current version before use.

Table 111. Test Limits

Input voltage (Vdc)	Input pin	Output voltage (We)	Output pin
-l5.0 ± 0.015	S	+15.0 ± 0.130	7
-13.0 ± 0.015	5	+11.267 ± 0.110	7
-10.0 ± 0.010	5	+ 8.667 ± 0.085	7
zero ± 0.010	5	Zero ± 0.050	7
-15.0 ± 0.015	6	+13.0 ± 0.130	7
-13.0 ± 0.015	6	+11.267 ± 0.110	7
-10.0 ± 0.010	6	+ 8.667 ± 0.085	7
zero ± 0.010	6	Zero ± 0.050	7
-13.0 ± 0.015	8	+13.0 ± 0.130	9
-10.0 + 0.010	8	+lO.0 + 0.100	9
zero ± 0.010	8	zero ± 0.050	9
+10.0 ± 0.010	8	-10.0 ± 0.100	9
+13.0 ± 0.015	8	-13.0 ± 0.130	9
-13.0 ± 0.015	26	+13.0 ± 0.130	19
-10.0 ± 0.010	26	+10.0 ± 0.100	19
zero ± 0.010	26	zero ± 0.050	19
+10.0 ± 0.010	26	-10.0 ± o.100	19
+13.0 ± 0.015	26	-13.0 ± 0.130	19
+13.0 ± 0.015	27	-13.0 ± 00.130	10
+10.0 ± O.010	27	- 7.640 ± 0.075	10
+10.0 ± O.010*	27	zero ± 0.050	10
zero ± 0.010	27	zero ± 0.050	10

•Pin 12 connected to +15.0 Vdc (PSI) this step only.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:38Z
Check the source to verify that this is the currentversion before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging and marking shall be in accordance with the level A requirements of Packaging Data Sheet (PDS), AMC form 1029, 10545705.

6. NOTES

6.1 The notes specified in the general specification MIL-C-50785 are applicable to this specification.

custodian

Army-MU

Preparing Activity

Army-m
Project No.1285-A315

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:38Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
Department of the Army

OFFICIAL BUSINESS

Commanding Officer
Frankford Arsenal
ATTN: SMUFA-J4000
Philadelphia, Pa. 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:38Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:38Z
Check the source to verify that this is the current version before use.